

AbangCebu AI — Authentication Baseline Latency & Load Strategy

Version: 1.0.0 | **Ticket:** SCRUM-72 | **Sprint:** 1 — Foundations & Core Infrastructure
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Status: ✔ Approved | **Tools:** k6 (Grafana) + Playwright + Supabase Dashboard

Performance Targets: Login P95 ≤ 500ms · Session Validation P95 ≤ 50ms · Registration P95 ≤ 600ms · Error Rate < 1%
Peak Load: 50 login req/s · 400 session validations/s · 800 concurrent sessions

1. Baseline Latency Thresholds

OPERATION	LAYER	P50	P95	P99
Session Validation	Edge Middleware	≤ 15ms	≤ 50ms	≤ 100ms
Silent Token Refresh	Middleware → GoTrue	≤ 80ms	≤ 150ms	≤ 300ms
Login Transaction	API Route → GoTrue → Profile	≤ 250ms	≤ 500ms	≤ 800ms
Registration Transaction	API Route → GoTrue → Profile	≤ 300ms	≤ 600ms	≤ 1000ms
Logout	API Route → Cookie clear	≤ 50ms	≤ 100ms	≤ 200ms
Password Reset	API Route → GoTrue → Email	≤ 200ms	≤ 400ms	≤ 600ms
OAuth Callback (Google)	Callback → Token exchange	≤ 400ms	≤ 700ms	≤ 1200ms

2. Traffic Projections (Month 3–6)

METRIC	LOW	MEDIUM	PEAK
Daily Active Users	500	2,000	8,000
Daily Logins	300	1,200	5,000
Daily Registrations	20	80	300
Peak Concurrent Sessions	50	200	800
Peak Login req/s	5	20	50
Peak Session Validations/s	25	100	400

3. Peak Traffic Scenarios

SCENARIO	LOAD MULTIPLIER	DURATION	TRIGGER
Enrollment Week (Jun/Aug)	3× normal	5–7 days	University enrollment, bedspace searches
BPO Shift Change (Daily)	2× normal	30–60 min	6AM/2PM/10PM shift transitions
Sinulog Festival Week	4× normal	7–10 days	Short-term rental surge
Viral Social Media Post	5–10× normal	2–4 hours	TikTok / FB group share

4. k6 Load Test Specifications

SCENARIO	VUS	DURATION	KEY THRESHOLDS
Steady-State Auth	20	9 min (2m ramp + 5m sustain + 2m down)	Login P95 < 500ms, error < 1%

Peak Burst	100–200	3 min (50 req/s peak)	Login P95 < 600ms, error < 5%
Middleware Stress	500–1000	3 min (400 req/s constant)	P95 < 100ms, error < 1%

5. Caching Strategy

LAYER	WHAT	TTL	IMPLEMENTATION
Edge Runtime	JWT decode result	Request-scoped	<code>jose.jwtVerify()</code>
GoTrue	Access token	3600s (1 hour)	Supabase Auth config
GoTrue	Refresh token	30 days	RTR with 30s grace
Browser	<code>sb-* -auth-token</code>	Session/30 days	<code>@supabase/ssr</code>
PostgreSQL	Profiles role	Pool TTL	Supavisor transaction mode

6. Connection Pool Recommendations

SETTING	DEV	STAGING	PRODUCTION
Pool Mode	Transaction	Transaction	Transaction
Pool Size	15	30	60
Max Client Connections	200	500	1000
Idle Timeout	60s	30s	20s
Statement Timeout	30s	10s	8s

7. Rate Limiting Strategy

ENDPOINT	WINDOW	LIMIT	KEY
<code>POST /api/auth/login</code>	15 min	5 per email	<code>email + IP</code>
<code>POST /api/auth/register</code>	1 hour	3 per IP	<code>IP</code>
<code>POST /api/auth/reset-password</code>	1 hour	3 per email	<code>email</code>
General API routes	1 min	60 per session	<code>session_id</code>

8. Pass/Fail Criteria Summary

METRIC	PASS	✓ FAIL ✗
Login P95	≤ 500ms	> 800ms
Login P99	≤ 800ms	> 1200ms
Session validation P95	≤ 50ms	> 150ms
Registration P95	≤ 600ms	> 1000ms
Error rate (steady)	< 1%	> 2%
Error rate (peak)	< 5%	> 10%
Connection pool usage	< 80%	> 95%

